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Math 215A

Homework 3
Due: October 14, 2026

Rings are understood to be commutative, unless stated otherwise.

Problem 1

Show that the kernel of the $\mathbb{C}$-algebra homomorphism $\mathbb{C}[x,y] \to \mathbb{C}[t]$ given by $x \mapsto t^2$ and $y \mapsto t^3$ is the ideal $(x^3 - y^2)$. (One possible approach is to show first that every element of the quotient ring $\mathbb{C}[x,y]/(x^3-y^2)$ can be written as $f(x) + g(x)y$ for some polynomials f and g.) Deduce that the ideal $(x^3 - y^2)$ in $\mathbb{C}[x,y]$ is prime.

Proof: Let $\varphi : \mathbb{C}[x,y] \to \mathbb{C}[t]$ be $\mathbb{C}$-algebra homomorphism described above. Note that $(x^3 - y^2) \subseteq \ker\varphi$. We want to show that $\ker\varphi \subseteq (x^3-y^2)$. Note that, $\mathbb{C}[x,y] \cong (\mathbb{C}[x])[y]$. Consider elements

$$f \in \frac{\mathbb{C}[x,y]}{(x^3-y^2)} \cong \frac{(\mathbb{C}[x])[y]}{(x^3-y^2)} = Q.$$

Note

$$f = \sum_{k=0}^{n} P_k y^k$$

for $P_k \in \mathbb{C}[x]$. However, note that $y^2 = x^3$ in Q. This means

$$y^k = y^{\lfloor \frac{k}{2} \rfloor * 2 + k \bmod 2} = x^{3 * \lfloor \frac{k}{2} \rfloor} y^{k \bmod 2} \in Q$$

Thus

$$\begin{aligned} f &= P_0 + P_1 y + \sum_{k=2}^{n} P_k x^{3*\lfloor \frac{k}{2} \rfloor} y^{\frac{1-(-1)^k}{2}} \\ &= \left(P_0 + \sum_{k=1}^{\lfloor \frac{n}{2} \rfloor} P_{2k} x^{3k} \right) + \left(P_1 + \sum_{k=1}^{\lfloor \frac{n-1}{2} \rfloor} P_{2k+1} x^{3k} \right) y \in Q \end{aligned}$$

Let

$$g = \left(P_0 + \sum_{k=1}^{\lfloor \frac{n}{2} \rfloor} P_{2k} x^{3k} \right) \in \mathbb{C}[x]$$

$$h = \left(P_1 + \sum_{k=1}^{\lfloor \frac{n-1}{2} \rfloor} P_{2k+1} x^{3k} \right) \in \mathbb{C}[x].$$

Thus $f = g + hy \in Q$. Note that since $(x^3 - y^2) \subseteq \ker\varphi$, φ factors through Q. Hence consider the induced map $\overline{\varphi} : Q \to \mathbb{C}[t]$. Suppose $f \in \ker(\overline{\varphi})$. Let $f = g + hy$ for $g, h \in \mathbb{C}[x]$.

$$\begin{aligned} 0 = \overline{\varphi}(f) &= \overline{\varphi}(g(x) + h(x)y) \\ &= g(t^2) + h(t^2)t^3. \end{aligned}$$

Thus

$$h(t^2)t^3 = -g(t^2).$$

Note the left hand side has odd degree, while the right has even degree if $h, g \neq 0$ which is a contradiction. Thus $h, g = 0$ and $f = 0$. Hence $\overline{\varphi}$ has kernel (0). Thus $\ker(\varphi) = (x^3 - y^2)$.

□

Problem 2

For any commutative ring R, show that $\mathrm{Spec}(R)$ is quasi-compact. (That is, if $\mathrm{Spec}(R)$ is the union of some collection of open subsets, then it is the union of finitely many of them. In point-set topology this would just be called "compact". The word "quasi-compact" is meant to emphasize that these topological spaces are not necessarily Hausdorff.)

Proof: A space X is quasi-compact if and only if the following holds: for any collection $\{C_\alpha\}_{\alpha \in \Lambda}$ of closed subsets of X with

$$\bigcap_{\alpha \in \Lambda} C_\alpha = \emptyset,$$

there exist $\{\alpha_1, ..., \alpha_n\} \subseteq \Lambda$ such that

$$\bigcap_{k=1}^{n} C_{\alpha_k} = \emptyset.$$

This formulation of quasi-compactness, comes from taking the standard open cover definition and taking complements. We will use this formulation, as closed sets have a nicer description in $\mathrm{Spec}(R)$. Every closed subset of $\mathrm{Spec}(R)$ is of the form $V(I) = \{\mathfrak{p} \in \mathrm{Spec}(R) : I \subseteq \mathfrak{p}\}$ for some ideal $I \subseteq R$. Thus any collection of closed sets corresponds to a collection of ideals $\{I_\alpha\}_{\alpha \in \Lambda}$ with $C_\alpha = V(I_\alpha)$. Note that

$$\bigcap_{\alpha \in \Lambda} V(I_\alpha) = V\left(\sum_{\alpha \in \Lambda} I_\alpha\right),$$

since a prime contains every I_α if and only if it contains their sum. Thus we want to show that for any collection of ideals $\{I_\alpha\}_{\alpha \in \Lambda}$ such that for all $\mathfrak{p} \in \mathrm{Spec}(R)$:

$$\sum_{\alpha \in \Lambda} I_\alpha \not\subseteq \mathfrak{p},$$

there exist $\{\alpha_1, ..., \alpha_n\} \subseteq \Lambda$ such that for all $\mathfrak{p} \in \mathrm{Spec}(R)$:

$$\sum_{k=1}^{n} I_{\alpha_k} \not\subseteq \mathfrak{p}.$$

This becomes trivial, because since

$$\sum_{\alpha\in\Lambda} I_\alpha \not\subseteq \mathfrak{p},$$

it cannot be in any prime ideal (and hence maximal ideal). Furthermore, it implies that it is not maximal itself and since every proper ideal is contained in a maximal ideal or is maximal itself, this implies

$$\sum_{\alpha\in\Lambda} I_\alpha = R \Rightarrow 1 \in \sum_{\alpha\in\Lambda} I_\alpha.$$

By the definition of sum, there exists $\{\alpha_1, ..., \alpha_n\} \in \Lambda$ such that

$$1 \in \sum_{k=1}^{n} I_{\alpha_k} \Rightarrow \sum_{k=1}^{n} I_{\alpha_k} = R \Rightarrow \sum_{k=1}^{n} I_{\alpha_k} \not\subseteq \mathfrak{p}$$

for any $\mathfrak{p} \in \operatorname{Spec}(R)$. □

Problem 3

Let R be a nonzero commutative ring. Let I and J be sets of different cardinalities. Show that the free R-modules $R^{\oplus I}$ and $R^{\oplus J}$ are not isomorphic. (Hint: this is true when R is a field.)

Proof: Note that R must have a maximal ideal of $\mathfrak{m}$. Note that

$$\frac{R^{\oplus I}}{\mathfrak{m}^{\oplus I}} \cong \left(\frac{R}{\mathfrak{m}}\right)^{\oplus I}.$$

Suppose $R^{\oplus I} \cong R^{\oplus J}$ as R modules, thus there exists a R-linear map $T : R^{\oplus I} \to R^{\oplus J}$. Thus we have the following commutative diagram

$$\begin{array}{ccc} R^{\oplus I} & \xrightarrow{\quad T \quad} & R^{\oplus J} \\ \Big\downarrow{\scriptstyle \pi_I} & & \Big\downarrow{\scriptstyle \pi_J} \\ (R/\mathfrak{m})^{\oplus I} & \xrightarrow[\tilde{T}]{} & (R/\mathfrak{m})^{\oplus J} \end{array}$$

$\tilde{T}$ is a $R/\mathfrak{m}$ isomorphism. Thus we have a isomorphism of vector spaces, hence $|I| = |J|$. □